23GE-1106- Engineering Chemistry
Session 2024-25 (odd Term) Section I

1. **Course Teacher** : **Dr. Megha Sawangikar**
   Office: SC0-09 Ground Floor, Department of Applied chemistry
   Office Hours : **Monday to Friday 10. 00A.M.to 5.00 P.M**
   Phone **9423119016**
   Email : **10419@ycce.in**
   Course Website :

2. **Course Objectives**
   1. To impart knowledge of the subject enriching students to understand the role of Chemistry in the field of engineering.
   2. To develop analytical capabilities of students for engineering materials
   3. To inculcate habit of scientific reasoning to do the task rationally.

3. **Course Outcomes-** At the end of the course, the student will be able to
   1. **Illustrate** qualitative and quantitative aspects of water for industrial and domestic applications. (L3)
   2. **Apply** concepts of electrochemistry for energy storage devices and corrosion. (L3)
   3. **Explain** basic principles of spectroscopy and its applications. (L3)
   4. **Establish** insight into advanced engineering materials. (L3)

   Mapped Program Outcomes:PO1,PO2,PO7,PO10,PO12

   PO1 Engineering knowledge: Apply the knowledge of mathematics, science, and an engineering fundamental, specialization to the solution of complex engineering problems
   PO2 Problem analysis: Identify, formulate, review research literature, and problems reaching substantiated analyze complex engineering conclusions using first and engineering sciences.
   PO7 Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
   PO10 Communication: Communicate effectively on complex engineering activities with the community engineering and with society at large, such as, being able to comprehend and write effective design documentation, make reports ad effective presentations, and give and receive clear instructions.
   PO12 Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

4. **Grading:** Grading is based on the following components that are weighted as described below

SN	Criteria	Distribution of marks(%)
1	Quizzes/Online tests	5
2	ASSESSMENT METHODS (Teacher's Assessment using ANY TWO INNOVATIVE methods))	12
3	Attendance% **Attendance% [91 to 100 % - 3 Marks, 81 – 90 % - 1½ Marks,75 – 80 % - 1 Mark, <75 % - Detention / 0**	3
4	Mid Semester Examinations-I	30
5	Mid Semester Examinations-II	
7	End Semester Examination	50
	TOTAL	100

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5. Syllabus and Mark Distribution Unit Wise

	Applied Chemistry		L=3	T=0	P=0	CREDITS = 3
EVALUATION SCHEME						
MSE – I, MSE – II		TA	ESE	TOTAL		ESE DURATION
15	15	20	50	100		3 Hrs (Offline)

Unit	Syllabus	MSE-I	MSE-II	ESE
I	**Water Chemistry** Introduction, Potable water quality parameters. Hardness, Types of hardness. Sterilization. Desalination of water by R.O. Softening of water by Zeolite process and Ion Exchange Process (principle, advantages, and limitations). Numerical based on Hardness and Zeolite process. Boiler trouble (Scale and sludge). Contemporary issues related to the topic.[07 Hrs.]	5	- -	8 OR 9
II	**Electrochemistry** Introduction, metallic and electrolytic conductance. Electrode and electrode potential. Nernst Equation, numericals and applications. Faraday's laws and numericals. Industrial applications: Electroplating, Electrolytic refining, Corrosion- Definition, Causes, theories of corrosion- dry, wet and differential aeration. Contemporary issues related to the topic [07 Hrs.]	5	- -	8 OR 9
III	**Energy storage device** Battery: Introduction, Characteristics, and general applications. Lithium-ion battery, Glass battery, H_2-O_2 Fuel cell. Differences between battery and a fuel cell. Recycling and safe disposal of batteries Supercapacitors: Definition, types, characteristics, and application. **H_2 as a green fuel:** Introduction, production, storage, and utilization. Contemporary issues related to the topic [06 Hrs.]	5	- -	8 OR 9
IV	**Drugs & Polymer chemistry** **Drugs:** Introduction, types of drugs, synthesis of commonly used drug molecules such as aspirin and paracetamol. **Polymer:** Introduction, Classification of polymers, Use and disposal of polymers. Properties of polymers - Solubility, Molecular Weight, Crystallinity, Glass transition temperature. Contemporary issues related to the topic [06 Hrs.]	- -	5	8 OR 9

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V	**Spectroscopic Techniques and Applications** Introduction, fundamentals, types, principles, and selection rules of spectroscopy. Basic principle and applications of UV- Visible, IR, NMR Spectroscopy and numericals. Contemporary issues related to the topic (06Hrs)		**5**	**8 OR 9**
VI	**Advanced Materials** **Nanomaterials:** Definition, Carbon Nanotubes and types. Applications of Nanomaterials in electronics, environment and medicine. **Chemical sensors:** Types and application. **Liquid Crystal Polymers:** Introduction, General properties and applications. **Polymers in electronic industries**: Introduction, Piezo, Pyroelectric, Ferroelectric polymers. **Smart materials:** Introduction, Properties and applications of Chromoactive, Photoactive and Magneto rheological materials. Contemporary issues related to the topic (07 Hrs.)		**5**	**8 OR 9**

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6. TEACHING PLAN

Lect. No.	Unit No.	*Topic Cover*	Date
1	I	Introduction to Engineering Chemistry syllabus, Handout	17/09/2024
2	I	Unit 1- Water Chemistry -Introduction, Potable water quality parameters Hardness, Types of hardness	18/09/2024
3	I	Numericals on hardness	19/09/2024
4	I	Softening of water by Zeolite process, Numericals	20/09/2024
5	I	Ion Exchange Process	24/09/2024
6	I	Sterilization, Desalination of water by R.O	25/09/2024
7	I	Boiler trouble (Scale and sludge).	26/09/2024
8	I	Class Test	27/09/2024
9	II	Introduction metallic and electrolytic conductance, Electrode and electrode potential..	04/10/2024
10	II	Nernst Equation, numericals and applications	08/10/2024
11	II	Faraday's laws and numericals.	09/10/2024
12	II	Industrial applications: Electroplating, Electrolytic refining	10/10/2024
13	II	Corrosion- Definition, Causes	11/10/2024
14	II	Theories of corrosion- dry, wet corrosion	15/10/2024
15	II	Differential aeration Corrosion	16/10/2024
16	II	Class Test	17/10/2024
17	III	Energy storage devices- Introduction, Characteristics, and general applications.	18/10/2024
18	III	Lithium-ion battery	22/10/2024
19	III	**ITL** Activity -Recycling and Safe Disposable of Batteries	23/10/2024
20	III	H_2-O_2 Fuel cell. Differences between battery and a fuel cell	24/10/2024
21	III	Glass Battery , Super capacitors: Definition, types, characteristics, and application	25/10/2024
22	III	**H_2 as a green fuel:** Introduction, production,.	05/11/2024
23	III	Storage, and utilization	06/11/2024
24	III	**Revision**	07/11/2024
25	III	Class Test	08/11/2024

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26	IV	**Drugs & Polymer chemistry** **Drugs:** Introduction, types of drugs,	19/11/2024
27	IV	Synthesis of commonly used drug molecules such as aspirin and paracetamol.	20/11/2024
28	IV	**Polymer:** Introduction, Classification of polymers,	21/11/2024
29	IV	Use and disposal of polymers.	22/11/2024
30	IV	Properties of polymers - Solubility, Molecular Weight	26/11/2024
31	IV	Crystallinity, Glass transition temperature	27/11/2024
32	IV	Revision /Class test	28/11/2024
33		**TA 1** Open Book Test	29/11/2024
34	V	**Spectroscopic Techniques and Applications** Introduction, fundamentals	03/12/2024
35	V	Types, principles, and selection rules of spectroscopy.	04/12/2024
36	V	Basic principle and applications of UV- Visible, Numericals	05/12/2024
37	V	Basic principle and applications of IR Spectroscopy	06/12/2024
38	V	Basic principle and applications of NMR Spectroscopy	10/12/2024
39	V	**CBS**-Microwave Spectroscopy	11/12/2024
40	V	Revision/Class Test	12/12/2024
41	VI	**Advanced Materials** **Nanomaterials:** Definition, Carbon Nanotubes and types.	13/12/2024
42	VI	Applications of Nanomaterials in electronics, environment and medicine.	17/12/2024
43	VI	**Chemical sensors:** Types and application	18/12/2024
44	VI	**Liquid Crystal Polymers:** Introduction, General properties and applications.	19/12/2024
45	VI	**Polymers in electronic industries**: Introduction, Piezo, Pyroelectric, Ferroelectric polymers.	20/12/2024
46	VI	**Smart materials:** Introduction ,properties and applications of Chromoactive materials	24/12/2024
47	VI	Introduction ,properties and applications of photoactive and magneto rheological materials	26/12/2024
48	VI	Class Test	27/12/2024

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7. Examinations:

1. **QUIZZES / ONLINE TESTS (05 MARKS)**: There will be a TWO proctored on campus online tests / Quizzes as per the guide lines of HoD preferably at the time of MSE examinations.

2. **TEACHERS ASSESSMENT (12 MARKS) :**AS GIVEN IN Teacher's Assessment Plan

3. **SCHEDULED EXAMINATIONS: There will be 3 scheduled exams during semester. There will be TWO Mid Sem Examinations (MSE) and shall be compulsory for evaluation of course assessment marks. The schedule of these examinations is given in the Academic Calendar for the session published by the Dean (Academic matters)**
4. MSE #1 : On the basis of Unit 1 , 2 and 3 of the syllabus
5. MSE #2 : On the basis of unit 4, 5 and 6 of the syllabus.

REMSE : In lieu of at the most one MSE in which student remained absent on the basis of All Unit 1, 2, 3, 4, 5 and 6 of the syllabus

ESE : On the basis of Units 1 to 6 of the syllabus

1. Advanced Topics Planned:

S.N.	Advanced Topics Planned
1	Microwave Spectroscopy

2. Innovative Teaching Method Planned:

S.N.	Innovative Teaching Method
1	Case Study- Removal and safe disposal of batteries

3. TA- Activities planned

S.N.	Innovative Teaching Method
1	Self-Video based learning on all units
2.	Open book test

4. Text books

1	S S. Dara, A Textbook of Engineering Chemistry, S. Chand & Co New Delhi. Eleventh Edition.
2	P.C. Jain and Monica Jain, Engineering Chemistry, Dhanpat Rai & sons New Delhi, Sixteenth Edition.
3	P. W. Atkins, Physical Chemistry, Oxford Publications, Eighth edition.
4	Y.R. Sharma, Elementary organic spectroscopy, S. Chand and company private limited.

5. Reference Books

1.	B.K.Sharma Krishna, Engineering Chemistry, Prakashan media private LTD. 1st Edition, 2014.
2.	CNR Rao, Chemistry of Advanced Materials, Willey Publications, 1993.
3.	Fred. Billmeyer Jr., A textbook of polymer science, Wiley India, 2nd Edition.
4.	Robert B Leighou , Chemistry of Engineering Materials ,Hill Book Company, Inc New York

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5.	C.N. Banwell ,Fundamentals of Molecular Spectroscopy ,Mc Graw hill education , 4th Edition
6.	William C. O'Mara, Robert B. Herring, Handbook of Semiconductor Silicon Technology ,Noyes Publications Park Ridge, NJ, USA.1st Edition.

6. YCCE e- library book links [ACCESSIBLE FROM COLLEGE CAMPUS]

1	http://103.152.199.179/YCCE/Suported%20file/Supprted%20file/SERIES%20WISE%20BOOKS/CHEMISTRY/

7. MOOCs Links and additional reading, learning, video material

1	https://www.youtube.com/watch?v=XTt3gXB0a84
2	https://www.youtube.com/watch?v=iihYXx79QiE
3	https://www.youtube.com/watch?v=JfJ7MIP9Dco
4	https://www.youtube.com/watch?v=L2VSOccUrSk
5	https://www.youtube.com/watch?v=p5pk4Um6lsk
6	https://www.youtube.com/watch?v=zVDMgoffmC0

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